Sibling Influence on Study Choice

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Abstract

This paper studies spillovers from the study choices of older siblings to the study choices of younger siblings. I use Dutch admission lotteries in medical schools as a source of exogenous variation. I find that 70% of siblings' common study choices are due to sibling spillovers, and the remaining 30% are due to common preferences. I argue that parental investment of income and time in children is likely the driving force of sibling spillovers.

Keywords: admission lotteries, siblings, study choice, the Netherlands

JEL Codes: I23, I24, J10

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1 Introduction

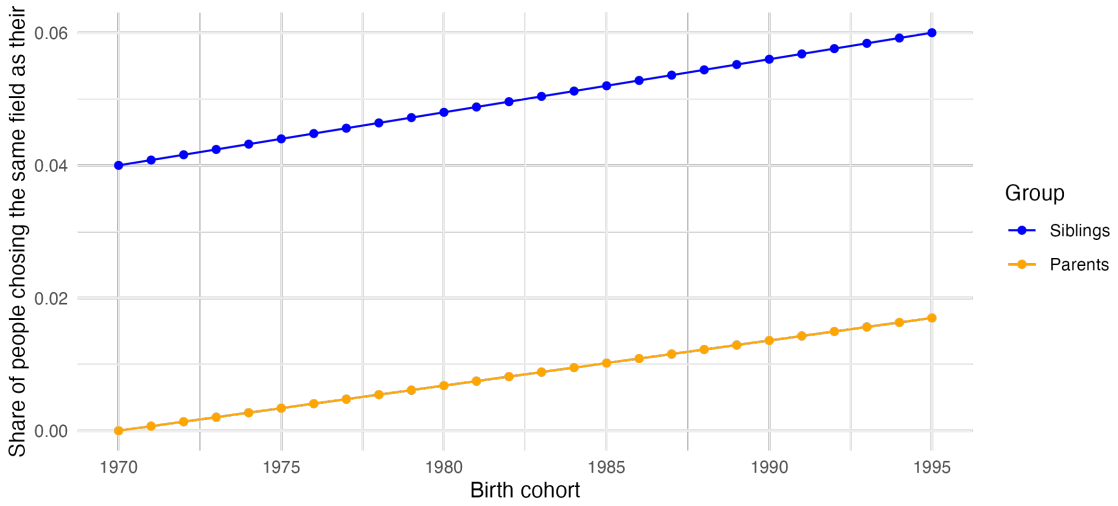
What to study in university? A person makes such a simple yet consequential decision early in life. The field of study choice affects earnings (Kirkeboen et al., 2016; Ketel et al., 2016; Altonji et al., 2012), partner choice (Kirkeboen et al., 2022; Artmann et al., 2021), and health outcomes (Leuven et al., 2013). Less is known about how a person chooses a field of study and, specifically, who influences this choice. Parents and siblings are members of the closest social network in most people's life. In the Netherlands, around 1% of children follow the field of study of their parents, and more than 5% that of their siblings (see Figure 1). Yet, siblings have been largely ignored in explaining study choice decisions. In 2021, almost 6000 published papers contained the words "parent" and "education", while there are less than 250 articles about "sibling" and "education" (see Figure 2). This study aims to fill this apparent gap.

This paper studies whether older siblings influence the study choice decisions of their younger siblings. Identifying peer effects is a challenging problem due to endogeneity, reflection problem, and correlated effects (Manski, 1993). I overcome these problems by exploiting a unique setting where admission to medical schools is determined by a lottery. I use administrative data on the universe of students who have ever participated in these lotteries. My identification strategy relies on the fact that admission to medical schools is random, hence, any differences between the study choice of younger siblings, whose older siblings are lottery winners or lottery losers can be attributed to the influence of the older sibling's lottery outcome.

I find that older siblings have a significant and economically meaningful influence on younger siblings' field of study choice. If an older sibling wins a lottery, a younger sibling is twice as likely to enroll in their older sibling's application program. A younger sibling is only 21.6% more likely to enroll in the same program as their older sibling is enrolled in. To further understand the nature of these common sibling study choices, I develop a simple theoretical model.

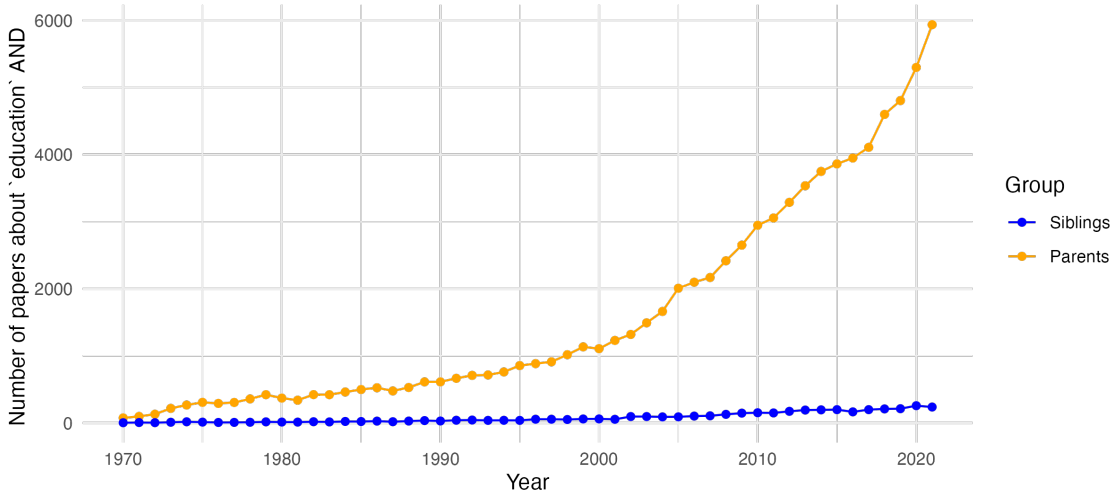
A model exploiting admission lotteries predicts competing effects of an older sibling's lottery outcome on a younger sibling's study choice. In the model, there are two types of programs: a lottery program (Medicine) and a program without a lottery (Economics). A lottery outcome is determined randomly, so there are no

Figure 1: The share of people who chose to study the same field as their siblings and parents (CLOSE APPROXIMATION OF REAL DATA)



Notes: The data comes from Statistics Netherlands. To calculate the share of people who chose the same field as their parents, I link individuals born in a specific year between 1970–1995 to their parents and calculate the mean of child–mother or child–father pairs who have ever been enrolled in the same field of study, i.e. a two–digit soicode. The share for siblings was calculated analogously.

Figure 2: The number of papers on "sibling and education" and "parent and education" in Scopus



Notes: The data comes from Scopus. To collect the data, I searched for papers congaing words 1) "education' AND "sibling"; 2) "education" AND "parent".

differences between winning and losing applicants. Assume that everyone applies to Medicine (which is the case in my setting). After a lottery is determined, winning applicants enroll in Medicine, and losing applicants enroll in Economics. If older siblings have a causal effect on their younger siblings, younger siblings would enroll in Medicine if their older siblings are lottery winners, and in Economics if they are lottery losers. However, if siblings simply share common educational preferences, younger siblings would enroll in Medicine regardless of the lottery outcome of their older siblings.

The highly stylized model makes key comparative predictions very clear. Under the pure sibling spillover effect, if an older sibling wins a lottery, it does not affect whether their younger sibling would enroll in the same program since younger siblings follow their winning older siblings as likely as their losing siblings. It does affect whether younger siblings would enroll in the older sibling's application program. Under the pure common preference effect, a model makes opposite predictions. Since younger siblings would choose a program their older siblings initially applied to, there is no effect of an older sibling's lottery outcome on whether a younger sibling enrolls in their older sibling's application program. There is, however, a positive effect on the younger sibling's enrollment in the same program as their older siblings.

To explore to what extent sibling common study choices in my setting are the result of causal sibling spillovers or common preferences, I make a simulation analysis of this theoretical model. The results of this study provide evidence that older siblings have a causal effect on their younger siblings, while siblings' study preferences are correlated only to a minor extent within a family. Specifically, 70% of common sibling's study choices are due to sibling spillovers, while the remaining 30% are due to common preferences.

This paper relates to a limited but growing amount of papers studying siblings spillovers in universities (Altmejd et al., 2021; Aguirrea and Matta, 2021; Goodman et al., 2015) and secondary schools (Dahl et al., 2021; de Gendre, 2021; Gurantz et al., 2020; van der Vleuten et al., 2020; Dustan, 2018; Joensen and Nielsen, 2018). The closest in spirit to this study is a paper by Altmejd et al. (2021). They study sibling spillover effects in college and major choices in Chile, Croatia, Sweden, and the United States. The authors use admission thresholds in the regression discontinuity design to compare younger siblings' study choices whose older siblings were

just above or below these cutoffs. They find that younger siblings follow their older siblings to the same college, but not to the same major. Similarly, Aguirrea and Matta (2021) exploit discontinuous admission cutoffs in the regression discontinuity design generated by Chile's centralized higher education system and find strong sibling spillover effects on university enrollment but not on the field of study choice.

The paper offers two main contributions to the literature. First, by exploiting the lottery design rather than the regression discontinuity design, this study estimates sibling spillover effects for a wider population. To estimate sibling spillovers, previous papers exploit admission cutoffs that generate quasi–random variation in enrollment probabilities. Hence, they identify effects among older siblings who are close to the admission margin, i.e. marginal applicants. To identify causal effects for a larger and perhaps more representative group of individuals, one can use a random lottery design that resembles an ideal experiment. In the paper, I use the unique institutional setting of admission lotteries in the Netherlands, which allows me to identify effects on inframarginal applicants.

Second, my findings shed light on the roots of differences in sibling spillovers. I argue that the family background is the main driver of heterogeneous sibling spillover effects. I show that parental investment in children explains large differences in spillover effects. Older siblings in poorer families have less influence on younger siblings' study choices. Siblings spillover effects are not existent in single–parent families. Taken together, these findings suggest that due to less parental investment of income and time in children, the quantity of sibling interactions decreases possibly due to lower quality of sibling's education advice. In addition, I show that older sibling's study experience matters in explaining sibling spillovers. I rule out studying together and role modeling as potential mechanisms by showing the absence of heterogeneous effects by the age difference, sex composition, and the ability of older siblings.

Understanding the nature of sibling spillovers matters for educational reforms. The common preference effects and the sibling spillover effects have two different policy implications. If sibling common study choices are due to common preferences, a policy maker cannot directly influence the decision of choosing a specific field. Common preferences are formed before the application process, which makes it hard to change one's decisions. If the common study choices are due to sibling spillovers,

programs and campaigns aimed at increasing enrollment in specific fields of study would have multiplicative effects. From a policy perspective, the results of the paper imply that additional information about the program can help in attracting more students into these fields. However, the policy recommendations should be applied with caution. Older siblings from poorer families and single–parent families show no influence on younger siblings' study choices. This raises a concern about whether strong family networks provide effective education support for the poor and single–parent families.

The paper is structured as follows. Section 2 describes the Dutch higher education and lottery systems. Section 3 presents a stylized model of study choice, testable predictions, and simulation analysis. Section 4 describes data. Section 5 presents my identification strategy. Section 6 presents the main results and possible mechanisms of sibling spillovers. Section 7 presents concluding remarks and discussion.

2 Institutional background

In the Netherlands, students who complete the academic track in secondary school can enroll in any Dutch university if their subject specialization in secondary school matches the chosen field of study. The majority of university programs have to accept all students, while some programs have quotas with a limited number of places, e.g. medical schools. 8 out of 13 universities provide medical training.

Until 1999, medical schools used a nationwide centralized lottery to admit students. The lottery was introduced as a response to the increasing number of applicants to medical schools which exceeded the number of available places. The Ministry of Education determines quotas. If the number of applicants exceeds the number of available places, a lottery determines who is admitted. The lottery is weighted, so students with a higher GPA have a higher probability of winning. A GPA is determined based on the grades from nationwide and externally graded high school exams, where six indicates a pass. Table 1 shows the weights and shares of groups of students in different lottery categories. For a more detailed description of Dutch admission lotteries and the educational system, see Leuven et al. (2013) and Ketel et al. (2016).

Table 1: Lottery categories

Category	GPA	Share	Weight
A	$GPA \geq 8.5$	0.02	2.00
B	$8.0 \leq GPA < 8.5$	0.05	1.50
C	$7.5 \leq GPA < 8.0$	0.09	1.25
D	$7.0 \leq GPA < 7.5$	0.21	1.00
E	$6.5 \leq GPA < 7.0$	0.22	0.80
F	$GPA < 6.5$	0.30	0.67
Other	–	0.11	1.00

Notes: A GPA is a grade point average on the final exams in high school. Share is the share of applicants in different categories that applied to the lotteries in the years 1988-1999. Weight indicates the relative probability of being admitted. The category "other" refers to students who did not participate in the nationwide high school exams, such as foreign students. This category is excluded from the analysis.

3 Predictions

3.1 Theory

There are multiple reasons why siblings follow similar educational pathways. First, siblings can have common educational preferences. Siblings are exposed to a similar environment, e.g. family, and they might attend the same school with similar teachers and classmates. As a result of such a close connection, it is quite common for siblings to choose a similar field of study not because siblings have a causal effect on each other, but due to a similar environment. This is called the common preference effect.

Second, an older sibling can have a direct influence on a younger sibling. An older sibling can provide valuable information about a field she is enrolled in. For example, an older sibling can inform a younger sibling about specific courses, the quality of peers, and the level of a program which can be helpful for a younger sibling. This type of information is hard to observe without studying in a program. Older siblings lower the costs of obtaining such information and make study options more salient to their younger siblings. Alternatively, older siblings can serve as role models, and younger siblings would simply imitate them regardless of the information provided.

This is called the sibling spillover effect.

To distinguish between common preference effects and sibling spillover effects, I use admission lotteries to medical schools. Admission lotteries allow me to estimate two treatment effects:

1. The effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in the program where an older sibling has *applied to*. Since medical schools are the only lottery program in this study, I call this an effect of the older sibling's lottery outcome on whether the younger sibling is enrolled in *medical school*.

2. The effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in the program where an older sibling is *enrolled in*. I call this effect an effect of the older sibling's lottery outcome on whether the younger sibling is enrolled in the *same program*.

Let us analyze a stylized example of a study choice model to predict these two treatment effects. Assume there are two study programs: medicine and economics. Medicine is a program that uses a lottery to admit students. There is a 50% probability of winning a lottery. Assume all lottery winners comply with the outcome of a lottery, i.e. full compliance. Economics is a program without a lottery, i.e. all students who apply are accepted.

First, let us consider a case under the common preference effects in Figure 3. Since all older siblings in the sample applied to a lottery, they revealed preferred medicine as their study choice. Under the common preference effect, the study preferences of younger siblings are identical among older siblings who are lottery winners and lottery losers. Hence, when the common preference effect is prevailing, the effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in a medical school is predicted to be *zero*, while the effect on enrollment in the same program is predicted to be *positive*.

Second, let us consider a case under the sibling spillover effect in Figure 4. Younger siblings follow their older siblings since they provide information about a study program or serve as role models. As a result, younger siblings are enrolling in the programs of their older siblings regardless of their lottery outcome. Hence, under the sibling spillover effect, the effect of an older sibling's study choice on whether a younger sibling is enrolled in the medical school is predicted to be *positive*, while the effect on enrollment in the same program is predicted to be *zero*.

Figure 3: A model of study choice under the common preference effect

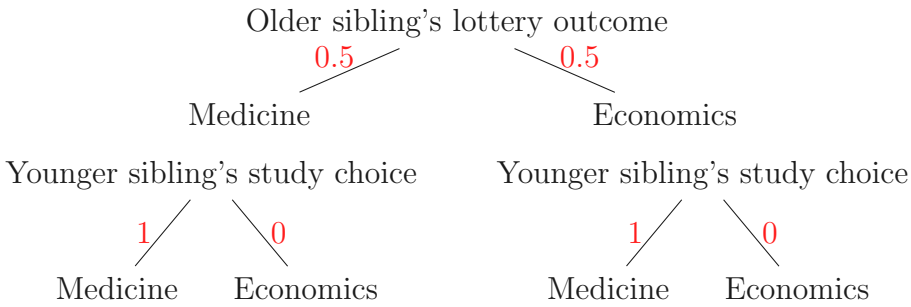
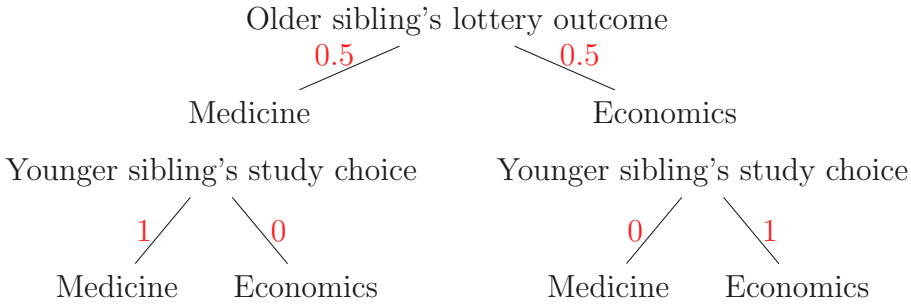


Figure 4: A model of study choice under the sibling spillover effect



The summary of model predictions is presented in Table 2. Under common preferences and sibling spillover models, the magnitude of treatment effects differs, which allows me to empirically test which effect is prevailing. These predictions are the main contribution to the literature as I can test directly to what extent sibling common study choices are explained by their common preferences or by the causal effect of older siblings.

Table 2: Predicted treatment effects of an older sibling's lottery outcome on a younger sibling's study choice

Outcome	Common preferences	Sibling spillovers
Δ_{MS}	0	+
Δ_{SP}	+	0

3.2 Model

This section provides results from a simulation analysis to determine the relative role of common preference effects and sibling spillover effects.

I follow the analysis of the stylized model described in Section 2. Note that the common preferences and sibling spillover effects lead to the same field of study choice of younger siblings whose older siblings are lottery winners: younger siblings always choose a medical school which is the program their older siblings are enrolled in. Since I am interested in disentangling the relative role of common preferences and sibling spillovers, I focus on simulating the study choice of younger siblings whose older siblings are lottery losers.

To simulate the fields of study choice of younger siblings, let me introduce some notations:

- Ω_{OS} – a set of older sibling's study programs.
- Ω_{-OS} – a set of study programs not included in Ω_{OS}.
- $p_i \in [0, 0.05, ..., 0.95, 1]$ – a probability that a younger sibling i chooses a program from a set Ω_{OS}.

Ultimately, p_i is the parameter determining the role of sibling spillovers, i.e. *sibling influence*. Hence, the *higher* the role of sibling influence, the *more* likely a younger sibling would choose a program an older sibling is enrolled in.

To estimate sibling spillovers, I use the following model:

$$Y_i = \delta D_i + \alpha_i + U_i \tag{1}$$

where Y_i indicates if in a sibling-pair i a younger sibling is enrolled in the 1) medical program and 2) same program as an older sibling; D_i indicates the result of the lottery of an older sibling; α_i are lottery category, year of the first lottery participation fixed effects, and their interaction.[1]

To empirically determine the role of the sibling influence in explaining siblings' common study choices:

[1]More details about the data can be found in Section 4.

1. I vary p_i between 0 and 1 in 0.05 increments.

2. Then with the probability p_i I draw a younger sibling's field of study program from a set Ω_{OS} and with the probability $1 - p_i$ – from a set Ω_{-OS}.

3. For each draw I estimate Model 1 to get reduced–form estimates of the older sibling's lottery outcome on the younger sibling's study choice.

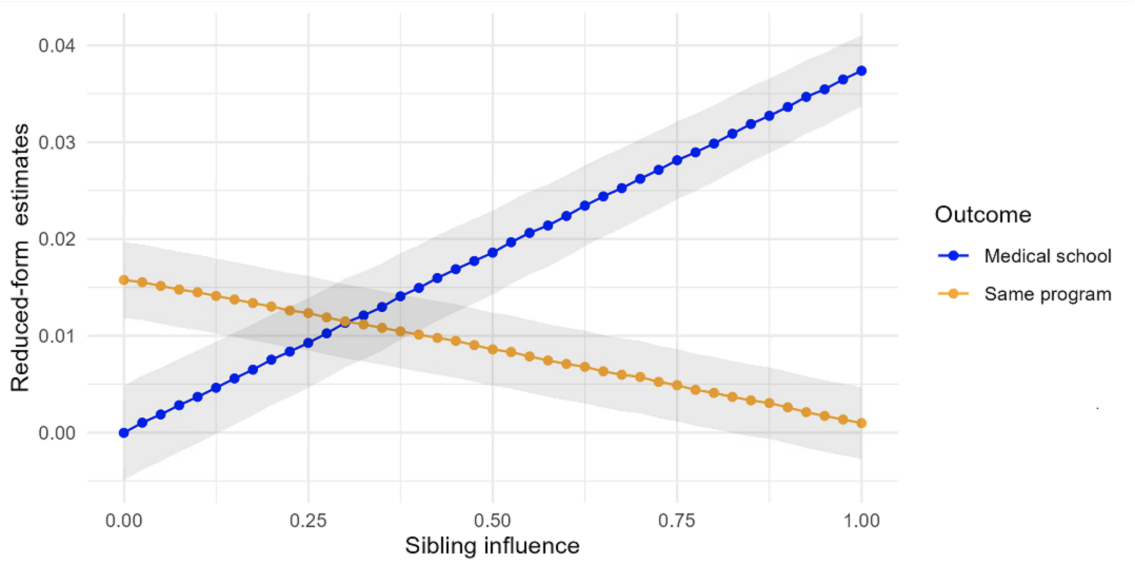
4. I repeat this process 1000 times for each p_i and take an average of point estimates and 95% confidence interval.

3.3 Simulation

The results of the simulation analysis are in Figure 5. The blue line shows reduced–form estimates of the effect of an older sibling's lottery outcome on a younger sibling's enrollment in a medical school. The yellow line shows reduced–form estimates of the effect of older siblings' lottery outcome on younger siblings' enrollment in the same program as their older siblings. The colored area is a 95% confidence interval.

As predicted by the stylized model, when the sibling spillover effect fully explains siblings' common study choices, i.e. $p_i = 1$, the effect of an older sibling's lottery outcome on whether a younger sibling is enrolled in a medical school is positive, i.e. $\Delta_{MS} = 0.037$ (s.e. $= 0.002$), while the effect on enrollment in the same program as an older sibling is zero, i.e. $\Delta_{SP} = 0.000$ (s.e. $= 0.002$). On the other hand, in the absence of the sibling spillover effect, i.e. $p_i = 0$, the effect of older sibling lottery outcome on whether a younger sibling is enrolled in a medical school is zero, i.e. $\Delta_{MS} = 0.000$ (s.e. $= 0.002$), while the effect on whether a younger sibling is enrolled in the same program as an older sibling is positive, i.e. $\Delta_{SP} = 0.016$ (s.e. $= 0.002$).

Figure 5: Reduced–form estimates of the effect of older sibling's lottery outcome of younger sibling's enrollment from 1000 simulations



Notes: The blue line shows reduced–form estimates of the effect of an older sibling's lottery outcome on a younger sibling's enrollment in a medical school. The yellow line shows reduced–form estimates of the effect of older siblings' lottery outcome on younger siblings' enrollment in the same program as their older siblings. The colored area is a 95% confidence interval. Sibling influence is a probability that a younger sibling chooses a program from a set of older sibling's study programs.

4 Data

4.1 Data source and sample

I use administrative data from the Statistics Netherlands (CBS) which allow linking all individuals to their parents and siblings. The registry contains information on different demographic and educational outcomes, i.e. age, sex, immigration background, university enrollment, field of study, and lottery outcomes. I observe lottery data on individuals who applied to medical schools between 1987 and 2004. For lottery participants, I observe the year when they participated in a lottery, lottery category, and lottery outcome.

To ensure that I observe first-year applicants, I exclude applicants who participated in the lottery in 1987 since there is no information about lottery participation in previous years. Since the lottery system was gradually abandoned from 1999

onward, I exclude lottery participants applying for the first time after this year. Applicants from category A are also excluded since almost all of them are admitted to a medical school. Further, I exclude applicants older than 20 at the moment they first participated in a lottery to exclude individuals who did not follow the usual educational track. Finally, I exclude applicants admitted via ministers' places since they do not participate in a lottery.

A unit of the analysis is a sibling pair. Hence, for every lottery participant, I link their younger siblings using a registry of their parents. I restrict the sample to lottery applicants who have at least one younger (half)–sibling. I include siblings who are born at least one year and not more than twenty years apart. In case there are several older siblings in a family participating in a lottery, I take the oldest older sibling to determine the treatment status of a younger sibling. The results are identical when I use the youngest older sibling, when I include families with only one older sibling participating in a lottery, or when I restrict the sample only to two sibling families.

4.2 Descriptive statistics

Table 3 shows the table of pre–determined characteristics of older and younger siblings by the outcome of the first lottery of older siblings. Around 60% of older siblings are female, 7% have a non–western background, and the average age at the first lottery is 18.8. The average number of siblings in a family is 3. Table 3 also represents the personal characteristics of younger siblings. Around 50% of the younger siblings are female, 8% have a non-western background, and the average age difference between younger and older siblings is 4.5 years. Around 50% of sibling pairs are same–sex pairs.

Tables A1 and A2 in the appendix present sample means and p–values separately by the lottery category for older and younger siblings. p–values are obtained from regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for lottery category, year of the first lottery participation, and their interaction. Personal characteristics of older and younger siblings are well balanced as indicated by p–values.

Table 4 presents descriptive statistics of enrollment of older and younger siblings.

Table 3: Personal characteristics of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers
Older siblings		
Female	0.59	0.59
Non-western immigrant	0.07	0.07
Age at first application	18.82	18.86
N	8029	10002
Younger siblings		
Female	0.49	0.48
Non-western immigrant	0.08	0.09
Age difference in years	4.50	4.43
N	12294	15348
Family level outcomes		
Same sex pairs	0.49	0.50
Family size	2.97	2.97
N	8029	10002

Notes: Since the lottery is weighted, the observed differences between lottery winners and losers in this table cannot be interpreted as causal.

Almost all older siblings are enrolled in a university: 99% among winners and 95% among losers. 95% of lottery winners enroll in a medical school, while only 44% of lottery losers do so. For younger siblings, around 82% are enrolled in a university, while 9% are enrolled in a medical school.

5 Empirical approach

I estimate the effect of an older sibling's lottery outcome on a younger sibling's study choice, using the following model:

$$Y_i = \delta D_i + X_i\beta + \alpha_i + U_i \tag{2}$$

where Y_i indicates if in a sibling-pair i a younger sibling is enrolled in the 1) medical program and 2) same program as an older sibling; D_i indicates the result of the first lottery of an older sibling; X_i includes indicators for siblings' sex and non-western origin, age of an older sibling at the first lottery; α_i are lottery category and year of

Table 4: Enrollment outcomes of older and younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers
Older siblings		
Enrolled in a university	0.99	0.95
Enrolled in a lottery program	0.95	0.44
N	8029	10002
Younger siblings		
Enrolled in a university	0.83	0.81
Enrolled in a lottery program	0.10	0.08
N	12294	15348

Notes: Since the lottery is weighted, the observed differences between lottery winners and losers in this table cannot be interpreted as causal.

the first lottery participation fixed effects as well as their interaction.

The effect of an older sibling's lottery outcome on a younger sibling's study choice is captured by δ. δ is the reduced–form estimate which captures the increased probability of a younger sibling enrolling in the program if an older sibling wins a lottery. Since the program admission of an older sibling is random conditional on lottery category, year of the first lottery participation fixed effects, and their interaction, the mean conditional independence assumption holds for the first lottery: $\mathbb{E}[U_i|X_i, \alpha_i, D_i] = \mathbb{E}[U_i|X_i, \alpha_i]$.

Admission lotteries help me to overcome the endogeneity problem as the variation in the older sibling's lottery outcome is exogenously determined. The reflection problem is eliminated because the study choice of the older sibling precedes and is assumed not to be influenced by the study choice of the younger sibling. I cluster standard errors at the family level, as treatment is assigned at this level and there can be multiple siblings within a family (Abadie et al., 2017).

6 Results

6.1 Main results

This section presents the main results of whether older siblings influence younger siblings' field of study choice.

Table 5 shows reduced–form estimates of sibling spillovers on the field of study choice. If an older sibling wins a lottery, a younger sibling's probability to be enrolled in a medical school increases by 3.2 percentage points. This doubles the share of siblings enrolled in a medical school compared to the control mean of 3.2%. A younger sibling is 1.2 percentage points more likely to be enrolled in the same program as an older sibling (5.6%). Based on the predictions derived earlier, these results indicate that the sibling spillover effect is stronger than the common preference effect.

In other words, younger siblings, whose older siblings lost a lottery, are less likely to enroll in a medical school and more likely to follow their older sibling's study program choice. This indicates that common educational preferences formed before the realization of the lottery outcome are not strong in explaining sibling common study choices. These findings indicate that older siblings have a direct causal effect on their younger siblings.

Table 5: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery	0.032***	0.012***
	(0.003)	(0.004)
Control mean	0.032	0.056
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. *p<0.1; **p<0.05; ***p<0.01

The reduced–form estimates of sibling spillovers found in the main analysis are as follows: $\Delta_{MS} = 0.032$ (s.e. $= 0.003$) and $\Delta_{SP} = 0.012$ (s.e. $= 0.004$). Comparing these estimates to the results of the simulation analysis in Figure 5, I find that 70%

of sibling common study choices are due to sibling spillovers and the remaining 30% are due to common preferences.

6.2 Mechanisms

This section presents suggestive evidence of mechanisms explaining sibling spillovers, i.e. studying together, role modeling, study experience, and parental investment.

Studying together

If siblings assign some value to study together with their older siblings, spillover effects should disappear if siblings are far apart in terms of age. To test this, Table A3 presents the results of sibling spillovers for siblings who are more than six years apart from each other, hence, they are unlikely to study together. The estimates show that sibling spillovers are not affected by the age distance between siblings. The fact that spillover effects do not disappear if there is more than six years difference between siblings indicates that sibling spillovers are not driven by the desire of younger siblings to study together with older siblings.

Role modeling

Older siblings can serve as role models to their younger siblings. It can be argued that the experience of siblings of the same sex is more relevant in choosing a field of study. If siblings serve as role models, there should also be stronger spillover effects from higher ability older siblings as they serve as a better example of academically successful students. Hence, younger siblings would be better off following their more successful older siblings than less successful ones. To explore role modeling as a possible channel, I look at the heterogeneity of sibling spillovers between same–sex and opposite–sex siblings as well as older siblings with higher and lower GPAs.

Table A4 presents the results of sibling spillovers by the sex composition of sibling pairs. Overall, there is not much systematic heterogeneity of sibling spillovers among siblings of the same and different sexes. Table A5 shows the results by the ability of older siblings. As a proxy of ability, I use the lottery category of an older sibling. Note, the higher the high school GPA of an older sibling, the higher the lottery

category. I interact the lottery outcome of an older sibling with an indicator for having a GPA higher than 7.5. Sibling spillovers are similar across older siblings of different ability levels. These findings provide suggestive evidence against siblings serving as role models.

Study experience

Younger siblings can take into account the experience of their older siblings at university. Ultimately, the older sibling's performance is a good proxy of how well the younger sibling would perform at university. If their siblings struggle in the program, younger siblings might shy away from a specific field due to its difficulty. Older siblings might decide to drop out of a program due to dissatisfaction with the content or social life. To explore whether older siblings' experiences matter, I explore heterogeneous spillover effects of the dropouts of older siblings. I use dropout as a proxy for the older sibling's experience, as it is an extreme form of negative study experience. Older siblings' dropout is not random, hence, the causal interpretation of these results is not guaranteed.

Table A6 presents the results of sibling spillovers by whether older siblings drop out from the program or stay. If older siblings drop out, the spillover effect is halved, i.e. younger siblings are less likely to enroll in their older sibling's application program. Reassuringly, if older siblings drop out, it does not affect the probability of younger siblings' enrollment in the same program. This result suggests that common educational preferences that are formed before enrollment, are unaffected by the negative experience of an older sibling during the study period. This indicates that we can, at least partly, interpret the smaller effect of sibling spillovers due to dropping out as a negative causal effect of study experience on study choices.

Parental investment

Parental investment in children depends on available parental resources, i.e. income and time. Poorer families have a lower income to invest in their children. Single parents have less time to devote to their children due to more stringent time constraints. Fewer available parental resources translate into less investment in children. Hence, the quality of children decrease in families with less available resources

(Becker and Tomes, 1976). The reduction in the quality of children decreases the benefits of sibling interactions. As a result, social interactions of siblings, such as asking for advice about education trajectories, are less likely to occur among families with fewer available resources.

To explore whether sibling spillovers are driven by the differences in parental investment in children, Table A7 investigates differences in sibling spillovers by parental income and Table A8 shows sibling spillovers in single– and two–parent families. Sibling spillovers are significantly lower among poorer families. Siblings' common preferences are also correlated to a lesser extent in poorer families. A stronger pattern is observed for single–parent families. Older siblings do not influence younger siblings' study choices in single–parent families nor their education preferences are correlated. These findings indicate that due to lower parental investment of income and time, sibling spillovers decrease.

In sum, sibling spillovers in the field of study choices do not differ by the age difference, sex composition, and the ability of older siblings. Spillover effects do vary by the quality of study experience of older siblings, parental income, and among single– and two–parent families. Ultimately, the analysis of heterogeneous effects is consistent with siblings providing valuable advice about study choices. Older siblings from better–endowed families provide higher quality advice about study trajectories. I rule out the studying together and role modeling channels as a possible explanation of sibling spillovers in study choice.

6.3 Robustness checks

Table A9 presents the results of sibling spillover effects using different sample definitions. In the main analysis, I use the oldest older sibling to determine the treatment status of all younger siblings, and these are baseline results in Column (1). In Column (2), I use the younger oldest sibling, in Column (3) – I use only families with one older sibling participating in a lottery, in Column (4) – only families with two siblings, and in Column (5), I do not put an upper bound on the age difference between siblings. Overall, the results are robust and stay virtually unchanged using different sample definitions.

7 Conclusion and discussion

The results of the paper provide evidence of strong sibling spillover effects in study choice. These findings have relevant policy implications. Policymakers can use information campaigns to better inform students of their study options, which would lead to multiplicative effects due to spillovers. However, if one aims at targeting disadvantaged students, i.e. students from poorer families or students from single–parent families, such programs can be less effective in achieving desired goals.

Interestingly, previous papers find no sibling spillovers on the field of study choice (Altmejd et al., 2021; Aguirrea and Matta, 2021). In contrast, I find substantial positive effects on the field of study choice. There are two possible explanations for these differences. First, previous papers studying sibling spillover effects use regression discontinuity design exploiting admission cutoffs. The estimated treatment effects are for younger siblings whose older siblings are very close to the cutoff points, i.e. marginally admitted students. This study uses a design exploiting admission lotteries. The estimated treatment effects are for younger siblings whose older siblings won the lottery, i.e. inframarginal students. Perhaps, younger siblings gain little from marginally accepted older siblings in terms of field of study choice, while they benefit a lot from inframarginal older siblings. Hence, one should be cautious in generalizing the existence and nature of sibling spillover effects in different settings.

Second, note that the average enrollment in universities in my sample is 80–90%, while in previous studies it is ranging from 15–18% in Chile and Croatia to 35–40% in Sweden and the United States. The pattern of these results suggests sibling spillovers might depend non–trivially on institutional characteristics. Speculatively, in settings with high enrollment in universities sibling spillovers might be less important for a decision to enroll in a university, since almost everyone is enrolled, and more important for what to study. Hence, further research is needed to make conclusions about how sibling spillovers depend on the features of higher education systems.

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Appendix

Table A1: Balancing table of personal characteristics of older siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers	p–value
Lottery category B			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category C			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category D			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category E			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category F			
Female			
Non-western immigrant			
Age at first application			
N			

Notes: The p–values in the final column are from the regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for lottery category, year of the first lottery participation, and their interaction.

Table A2: Balancing table of personal characteristics of younger siblings by the outcome of the first lottery of older siblings

Outcome	Lottery winners	Lottery losers	p–value
Lottery category B			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category C			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category D			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category E			
Female			
Non-western immigrant			
Age at first application			
N			
Lottery category F			
Female			
Non-western immigrant			
Age at first application			
N			

Notes: The p–values in the final column are from the regressing personal characteristics on an indicator function for an older sibling winning the first lottery, including fixed effects for lottery category, year of the first lottery participation, and their interaction.

Table A3: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by the age difference

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Age difference > 6	()	()
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether siblings are six of more years apart. *p<0.1; **p<0.05; ***p<0.01

Table A4: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by the sex composition

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Same sex	()	()
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether siblings are of the same sex. *p<0.1; **p<0.05; ***p<0.01

Table A5: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by ability of older siblings

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × GPA > 7.5	()	()
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether older sibling's lottery category is B or C, i.e. GPA is higher than 7.5. *p<0.1; **p<0.05; ***p<0.01

Table A6: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by drop outs among older siblings

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Dropped out	()	()
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether older sibling dropped out. *p<0.1; **p<0.05; ***p<0.01

Table A7: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by parental income

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Parental income < median	()	()
N	27541	27541

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether parental income is lower than the median parental income. *p<0.1; **p<0.05; ***p<0.01

Table A8: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's study choice by single– and two–parents families

Outcome	Younger sibling is enrolled in the	
	Medical school	Same program
Older sibling wins a lottery		
	()	()
Older sibling wins a lottery × Single–parent family	()	()
N	27642	27642

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. In addition they include interactions with an indicator for whether siblings live in single–parents families, when younger siblings are 12–18 years old. *p<0.1; **p<0.05; ***p<0.01

Table A9: Reduced–form estimates of the effect of older sibling's lottery outcome on younger sibling's enrollment with different sample definitions

	Outcome	Oldest OS	Youngest OS	One OS	Two siblings	No age limit
1	Medical school	0.032***	0.031***	0.029***	0.030***	0.031***
		(0.003)	(0.003)	(0.003)	(0.006)	(0.003)
	Control mean	0.032	0.032	0.031	0.026	0.032
2	Same program	0.012***	0.012***	0.010***	0.010	0.012***
		(0.004)	(0.003)	(0.003)	(0.008)	(0.003)
	Control mean	0.056	0.056	0.055	0.059	0.055
	N	27642	27642	26646	7195	28222

Notes: Standard errors are clustered at the family level. All regressions control for siblings' sex, non-western origin, age of an older sibling at the first lottery, fixed effects for lottery category, year of the first lottery participation, and their interaction. Oldest OS – I use the oldest older sibling in a family to define a treatment status of younger siblings, Youngest OS – youngest older siblings, One OS – I use only families with one older sibling participating in a lottery, Two siblings – only families with two siblings, and No age limit – I do not put an upper bound on the age difference between siblings. *p<0.1; **p<0.05; ***p<0.01